

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Surds

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SURDS · CH1 · L10

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The Map

This lesson collapses Surds into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Simplify by square factors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Add and subtract like surds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Expand surd brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Rationalise the denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Simplify by square factors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q23

The volume is

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

Simplify surd

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

Finish: $2 + \sqrt{5}$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

23 The diagram shows a cuboid with a square cross section. x cm x cm $(25 - 3)$ cm
Diagram NOT accurately drawn The volume of the cuboid is $(13 + 65)$ cm³ Without using a calculator, find the value of x Give your answer in the form $a + b$ where a and b are integers. Show your working clearly. $x =$

YOUR SOLUTION

02 Add and subtract like surds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Expand surd brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q18

Simplify surd

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

Simplify surd

$$y\sqrt{x} + 21$$

Finish: $x = 19, y = 3$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

18 Given that $(8 - x)(5 + x) = yx + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly. $x = y =$

YOUR SOLUTION

04 Rationalise the denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

20 The area of a rectangle is 18 cm^2 . The length of the rectangle is $7\frac{1}{2} \text{ cm}$. Without using a calculator and showing each stage of your working, find the width of the rectangle. Give your answer in the form $a + b + c$ where a , b and c are integers. cm

YOUR SOLUTION

05 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2(2\sqrt{5} - 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q23

The cuboid has square cross

$$x^2(2\sqrt{5} - 3)$$

Given

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

Finish: $2 + \sqrt{5}$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

23 The diagram shows a cuboid with a square cross section. x cm x cm $(25 - 3)$ cm
Diagram NOT accurately drawn The volume of the cuboid is $(13 + 65)$ cm³ Without using a calculator, find the value of x Give your answer in the form $a + b$ where a and b are integers. Show your working clearly. $x =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Simplify by square factors
"work out", "show", "hence", a familiar exam phrase	Add and subtract like surds
"work out", "show", "hence", a familiar exam phrase	Expand surd brackets
"work out", "show", "hence", a familiar exam phrase	Rationalise the denominator
"work out", "show", "hence", a familiar exam phrase	Simplify surd
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.